

Network Security & Basic Traffic Monitoring Report

Introduction

Network security is the practice of protecting networks, systems, and data from cyber threats and unauthorized access. Understanding how data travels across networks is essential for identifying and preventing security risks.

Objectives

- Learn basic networking concepts.
- Understand common network attacks.
- Monitor network traffic using Wireshark.
- Analyze common protocols safely and ethically.

Tools Used

- Wireshark
- Command Prompt
- Web Browser

Practical Activity

A packet capture session was started using Wireshark. During the capture, websites were visited and a ping test was performed using:

```
ping google.com
```

Protocol filters were applied to identify and analyze different types of traffic.

Observations

DNS

DNS requests and responses were observed when accessing websites.

HTTPS/TLS

Encrypted TLS traffic was observed, demonstrating secure communication.

ICMP

ICMP Echo Requests and Echo Replies were observed during the ping test.

TCP

TCP packets showed connection establishment and communication between devices.

Ethical Considerations

All traffic monitoring was performed on my own system for educational purposes only. No unauthorized access or malicious activities were conducted.

Learning Outcomes

- Understood network security fundamentals.
- Learned the role of DNS, HTTP, HTTPS, ICMP, and TCP.
- Gained practical experience with Wireshark.
- Learned basic packet analysis techniques.
- Understood ethical guidelines for traffic monitoring.

Conclusion

The task successfully demonstrated how network communication occurs and how traffic can be monitored and analyzed using Wireshark. This activity improved my understanding of networking concepts and cybersecurity practices.